

Project Management Plan

Upstream Scenario
Modelling Framework

Finance



Project Definition, Objectives, and Scope

Objective: Develop and evaluate a controlled, transparent and reproducible framework for constructing and analysing alternative Upstream portfolio scenarios.

Business Need

Portfolio scenario analysis relied heavily on repeated manual spreadsheet manipulation, making alternative cases time-consuming to construct and making reproducibility, transparency and interrogation more difficult.

Project Response

Design and implement a controlled scenario modelling framework combining governed enterprise and manual planning data, an Excel user interface, and a modular Python analytical engine. Users can select modelling cases and apply reusable scenario operations while retaining traceability to the underlying planning data.

Intended Business Value

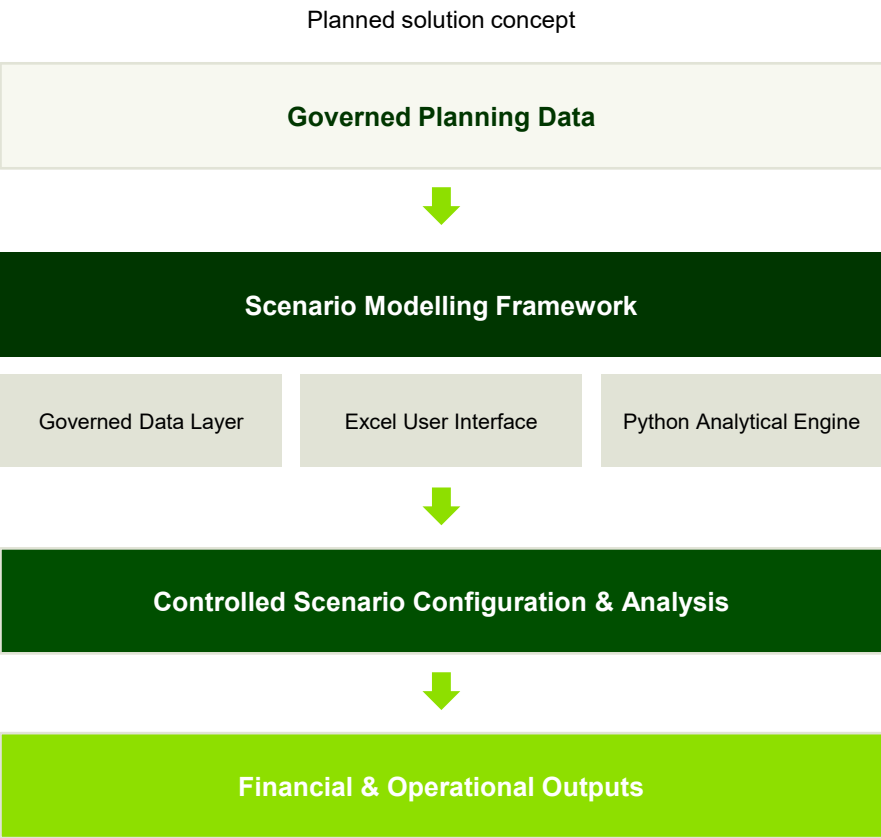
Improve optionality, transparency, repeatability and the ability to interrogate alternative portfolio scenarios, while preserving the role of expert business judgement.

In Scope

- Governed integration of planning data
- Financial Entity to modelling-case translation
- Controlled manual planning-data integration
- Scenario configuration and reusable transformation operations
- Financial and operational scenario outputs
- Validation, reporting and user controls

Deliberately Outside Current Scope

- Predictive or machine-learning forecasting
- Fully industrialised enterprise deployment
- Commodity-price functionality without a validated business methodology
- Fully validated country-specific fiscal/tax modelling
- Portfolio optimisation as an implemented production capability



Indicative solution boundary as defined at project outset; not an implemented system.

Project Delivery Timeline and Workstreams

Structured delivery across business support, academic outputs, and parallel application development

Phased delivery — Defined phases from discovery and design through build, validation and final evaluation.

Parallel workstreams — Business support, DT603 academic delivery and Streamlit development progress in parallel.

Milestone-led control — Progression governed by validation gates and formal submission checkpoints.

	1 Discovery and re-engagement Weeks 1–2	2 Terms of Reference and scope Week 3	3 Baseline data and mapping Weeks 3–4	4 Scenario build and configuration Weeks 4–6	5 Pilot evaluation and SME validation Weeks 6–8	6 Final evaluation and recommendation Weeks 8–10
 Business / LTP process support	- Review earlier prototype work - Re-engage Upstream portfolio stakeholders - Confirm business need	- Refresh terms of reference - Agree SME user group - Prioritise scenario operations	- Prepare trusted baseline dataset - Agree approach to financial entity aggregation	Build, test, and validate priority scenario operations	- Run alongside live LTP business process - Reconcile to BP referenced views	- Assess usability, traceability, scalability - Document limitations - Recommend future development path
 DT603 academic delivery	Confirm DT603 scope, outcomes, and delivery approach	- Create project management plan - Set-up repository - Set-up meeting cadence	Capture extraction, profiling, and architecture evidence	Develop core working data analysis solution	- Capture business validation evidence - Conduct stakeholder walkthroughs - Gather feedback	Complete DT603 submission package
 Parallel Streamlit app development	- Environment set-up and tooling review - Gather app specific requirements	- Set-up VS code workspace - Learn agentic development approach	Build Streamlit scaffold, navigation, and scenario input forms	- Connect the Streamlit app to the Python scenario engine - Add charts and comparison views	- Demo the app - Refine UI/UX - Capture feedback	Recommend next phase development path
 Milestones	Context & feasibility confirmed	Terms of Reference approved	Baseline & mapping approach agreed	Priority scenarios configured	Validation evidence captured	Final submission and recommendation

Integrated delivery timeline showing project phases, workstreams, milestones, and expected outputs

Data Acquisition and Transformation Plan

Progress tracker for the data sourcing, ingestion, transformation and reconciliation activities supporting Long Term Planning

#	Activity	Description
1	Pull SAP HANA data and reconcile to AFO grid	Extract data via SAP HANA dataflow and validate outputs against the AFO planning grid to ensure alignment, completeness and traceability of key financial measures.
2	Pull ADI ingested data and reconcile to AFO grid	Ingest data through ADL pipelines and perform reconciliation to the AFO grid to confirm consistency with SAP-sourced outputs and identify any transformation discrepancies.
3	Load RPE data into PRIX and reapply Capex Book Case tagging	Import RPE datasets into the O&G profile PBIX and recast Capex Book Case tag assignments using the current logic to align with existing modelling approach.
4	Update Capex Book Case tagging logic and re-run	Incorporate agreed changes to tagging rules based on business input, update transformation logic, and re-run outputs to ensure consistency and accuracy.
5	Develop modelling decision tables and gather input	Prepare structured decision tables (e.g. yes/no per case tag) and validate assumptions with stakeholders to support transparent and repeatable modelling logic.
6	Build reference and master data structure for modelling	Build reference tables to support consistent slicing of data (e.g. country, region, BP/OBO, asset hierarchy), aligned to business reporting structures and enable consistent scenario outputs.
7	Identify and assess tagging limitations	Review scenarios where case tagging cannot be reliably applied (e.g. commingled financial entities) and document constraints, impacts, and potential workarounds.
8	Compute and validate modelling values	Derive required rates (e.g. unit costs, conversion factors, allocation drivers) from reconciled datasets and validate against existing assumptions to ensure consistency for scenario modelling.
9	Replace source data with curated offline scenarios	Identify cases where base datasets need to be overridden and substitute with curated offline scenario inputs to reflect updated assumptions or alternative business views.
10	Validate scenario-ready dataset	Perform final checks across base data, derived rates and scenario inputs to ensure completeness, traceability and readiness for modelling use.
11	Create FE to Modelling case mapping	Prepare FE to modelling case tag mapping table to be applied to FIN cube data, then blend with the reference data.
12	Obtain effective tax rates	Request tax rates by country.

Data sourcing and transformation activity log – supporting detail to the delivery workstreams.

Plan Scope

This plan covers the end-to-end data flow from source systems (e.g. SAP HANA, ADL ingestion pipelines and BPE datasets) through to aligned and model-ready outputs used in Power BI and planning tools.

- Extraction of financial and operational data via defined dataflows and ingestion routes
- Reconciliation of data between source systems and the AFO planning grid to ensure consistency and traceability
- Transformation and enrichment logic, including application of Capex Book Case tagging and associated business rules
- Iterative updates to modelling logic based on validation input and evolving business requirements
- Identification and management of data gaps or constraints (e.g. commingled financial entities where tagging is not feasible)
- Preparation of decision tables to support transparent and repeatable modelling scenarios

Objective

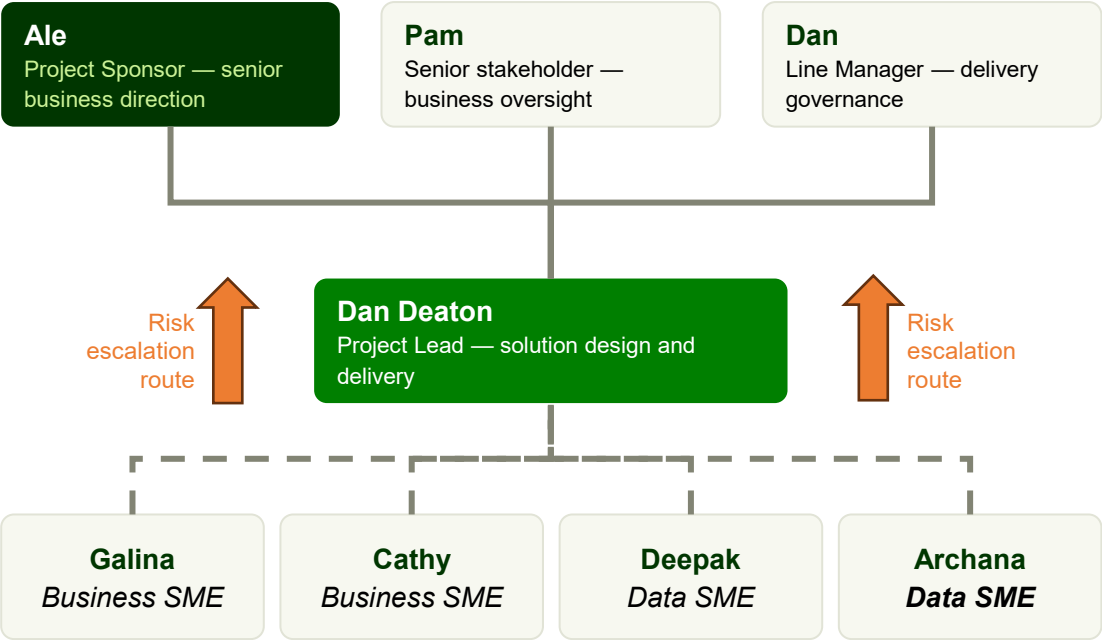
The objective is to ensure that all data used within LTP modelling is accurate, auditable, and aligned across systems, enabling consistent decision-making and reducing manual reconciliation effort.

Activities are tracked and reviewed through the delivery workstream, with ownership and progress maintained against each item.

Governance, Stakeholders and Delivery Control

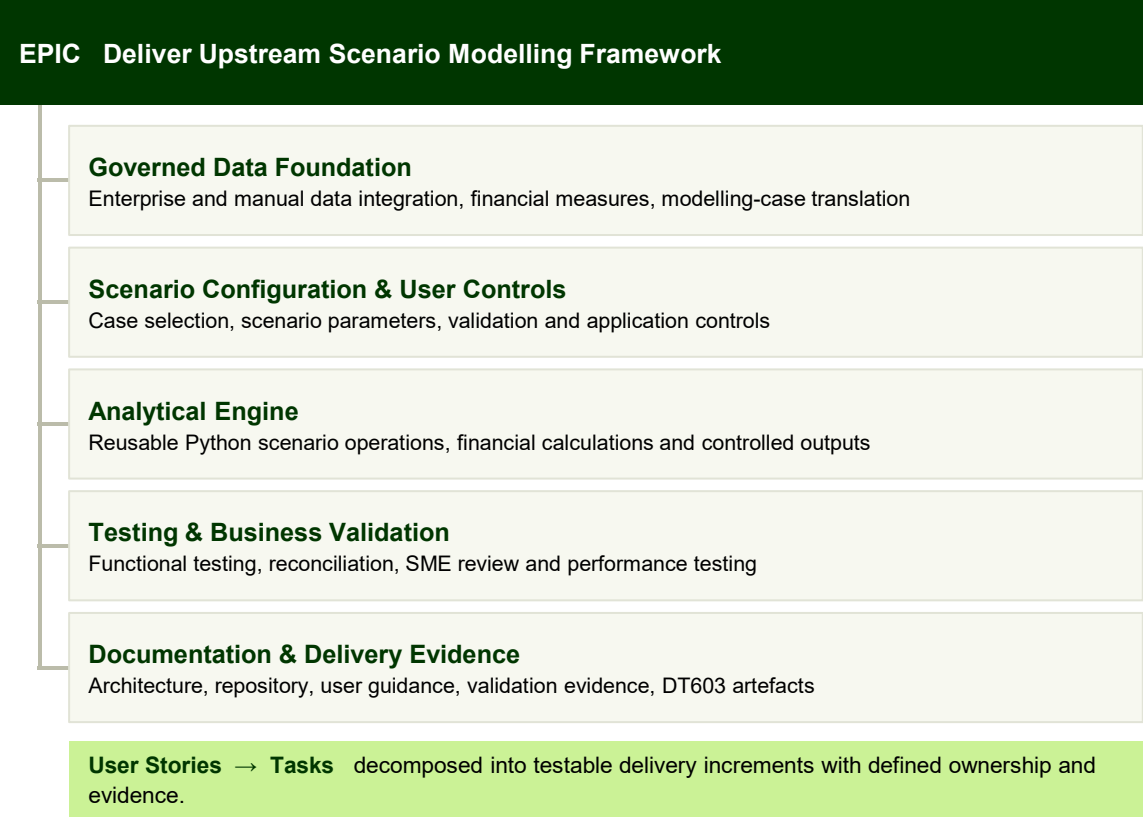
Planned governance structure, delivery accountabilities and structured decomposition of work

Project Governance



Subject Matter Experts — Galina, Calvin, Deepak, and Archana provide requirements clarification, data and methodology expertise, validation and challenge.

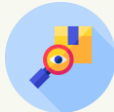
Structured Delivery Control



**Governance and Communication Controls**

**SME engagement**
direction, success criteria and key decisions

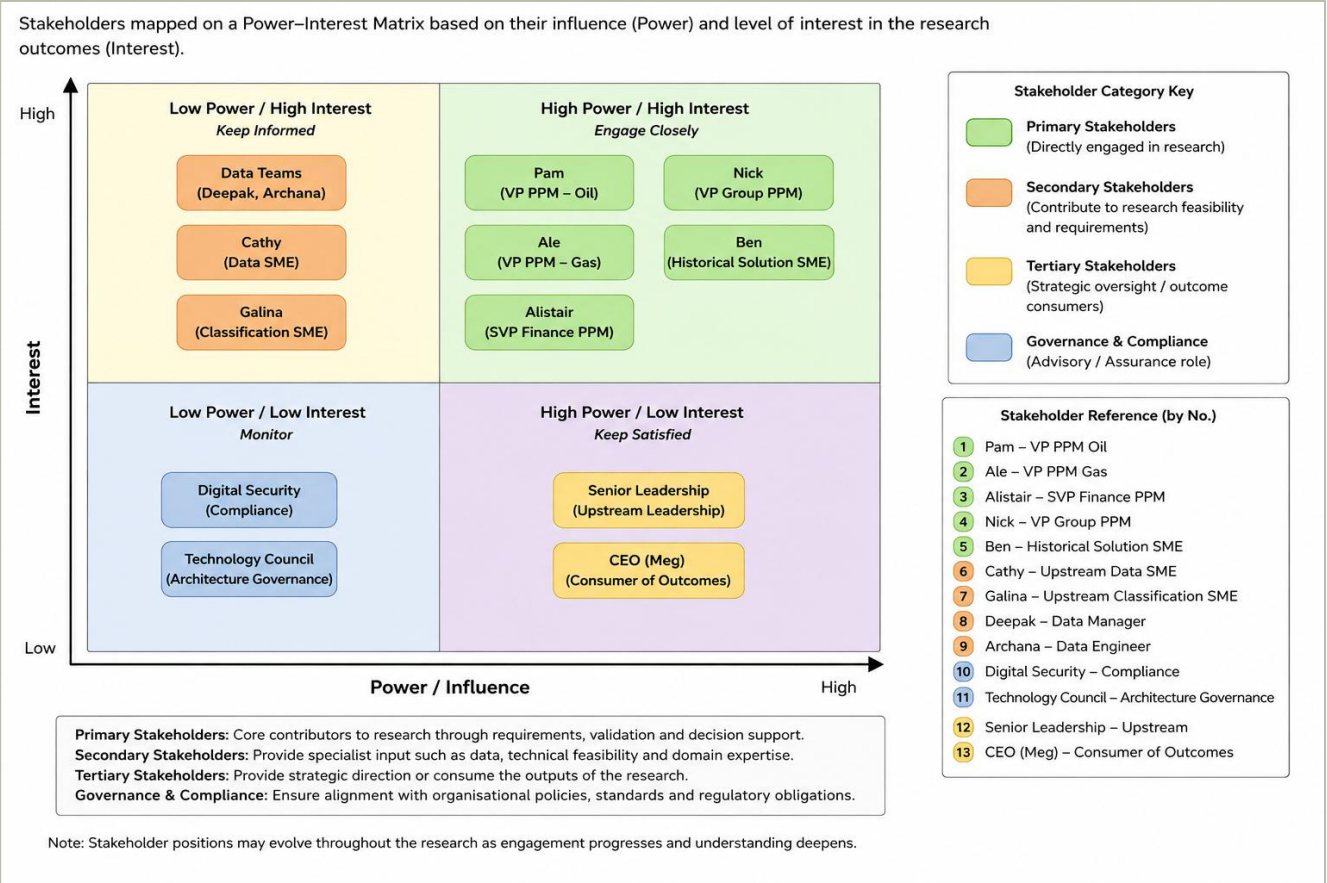
**SME engagement**
requirements, business rules, data understanding, validation

**Delivery tracking**
prioritised work decomposition and progress control

**Communication & assurance**
planned reviews, walkthroughs and evidence capture

Stakeholder Engagement and Communication Strategy

Mapping stakeholder influence and interest to plan proportionate engagement



Stakeholder power–interest matrix – stakeholders positioned by influence over, and interest in, the project.

Stakeholder Mapping and Analysis

- Assess stakeholders by relative power and interest
- Define the appropriate level of engagement
- Focus attention on those with the greatest influence on project direction, requirements, validation, and delivery

Engagement Classification

Primary Sponsor, senior business stakeholders and line management – managed closely

Secondary Subject matter experts and delivery contributors – kept satisfied and consulted

Tertiary Wider planning community and governance functions – kept informed

Critical Engagement

Sponsor and senior stakeholder involvement secures direction, prioritisation and access to governed data and expertise

Engagement intensity is planned according to stakeholder influence, interest and contribution.

Risk, Dependencies and Mitigation

Planned controls for the principal threats to successful project delivery

#	Threat	Gap	Category	L	I	Rating	Mitigation
1	Misinterpretation of stakeholder requirements	Capability	Technical	H	H	H	Structured operation definitions and stakeholder validation
2	Fragmented planning and financial data	Capability	Technical	H	H	H	Data profiling, reconciliation and standardised transformation rules
3	Financial logic validation challenges	Governance	Technical	M	H	H	SME review, testing and reconciliation against existing outputs
4	Organisational change and shifting priorities	Capability	Org	M	H	H	Iterative requirements review and stakeholder engagement
5	Competing initiatives (System of Performance, Horizon X, Lighthouse)	Capability	Org	M	M	M	Scope management and sponsor alignment
6	Mishandling commercially sensitive planning information	Governance	Ethical / Legal	L	H	M	bp governance controls and data classification policies
7	Researcher bias arising from insider position	Governance	Research	L	M	M	Triangulation and stakeholder validation
8	Dependency on specialist knowledge	Sustainability	Org	M	H	H	Knowledge transfer, documentation and transparent logic
9	Framework perceived as overly complex	Sustainability	Adoption	M	M	M	User-centred design, training, documentation
10	Prototype-led development evolves into enterprise platform expectations	Capability	Delivery	M	H	H	Maintain clear scope boundaries and governance process for enhancements

Risk Management Approach

Identify Threats captured across capability, governance, organisational and delivery dimensions.

Assess Each risk scored for likelihood and impact on a 5×5 matrix to set priority.

Mitigate Proportionate controls agreed and built into the delivery plan and ways of working.

Monitor Risks reviewed through project governance, with escalation to the sponsor where required.

Key Project Dependencies

- Timely access to governed planning and financial data
- Subject matter expert availability for review and validation
- Continued sponsor and line management support
- Approved bp environments for sensitive planning data
- Scope stability alongside competing initiatives

Project risk log – threats assessed by likelihood and impact, with planned mitigation. Ratings shown are as assessed at planning stage.

Risk control is integrated into delivery planning, with material risks assessed, mitigated and monitored through project governance.

Success Criteria, Validation and Project Closure

The project will be evaluated across the following dimensions to assess both the technical performance of the prototype and the value it delivers to the wider planning process.

Evaluation Dimension	Objective	Measure / Evidence
① Technical Capability	Demonstrate that the prototype reliably performs the intended scenario modelling functions.	Successful execution of scenario operations, validated data pipeline, repeatable outputs and reconciliation to trusted planning artefacts. Ability to configure and compare multiple portfolio scenarios, including manual inputs, portfolio construction and optimisation capability.
② Business Value	Improve planning efficiency, repeatability and decision support.	Estimated reduction in manual effort, increased consistency of scenario construction and SME assessment of planning usefulness.
③ Governance & Transparency	Demonstrate trusted and auditable analytical processes.	Documented assumptions, mappings, transformation rules, validation evidence, controlled manual inputs and security considerations.
④ Adoption & Usability	Confirm that planners can understand, operate and challenge the prototype.	Walkthrough feedback, usability observations, review comments and enhancement requests.
⑤ Strategic Contribution	Assess the prototype's contribution beyond the immediate use case.	Evidence of reusable architecture, lessons learned and contribution to wider initiatives (e.g. Horizon X).
⑥ Sustainability	Assess whether the approach can be maintained, extended or scaled.	Recommendations for future development, known limitations and suitability for tactical or enterprise implementation